



MASTER OF SCIENCE – DATA SCIENCE LEARNING PLAN

“I am deliberate and
afraid of nothing.”

Audre Lorde, Writer and Professor

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DEGREE PROGRAM SUMMARY

The MS Data Analytics degree prepares a diverse range of professionals for thriving careers in the dynamic field of data analytics. By arming graduates with the competencies needed to tackle business challenges through data mining, predictive analysis, analytics deployment, and compelling data storytelling techniques, the MSDA ensures they are well-prepared for success. Emphasizing both theory and practical application, the curriculum fosters the development of skills necessary to drive impactful change within organizations spanning various industries and sectors. The Data Science concentration enhances mastery in statistical and programming methodologies, delving into the advanced topics advanced analytics professionals will need such as machine learning, neural networks, and numerical optimization. The program includes a concentration-specific capstone, providing an opportunity for students to showcase their skills in a comprehensive manner

DEGREE PROGRAM OVERVIEW

WGU's data analytics master's degree program features a data science specialization that allows you to gain specific experience and skills in the data science field, preparing you for an exciting future. **Machine learning, advanced analytics, and numerical optimization** are critical additions to the data science concentration, helping students move forward with their career aspirations.

Course Code	Course Title	Credits
D596	The Data Analytics Journey	2
D597	Data Management	3
D598	Analytics Programming	3
D599	Data Preparation and Exploration	3
D600	Statistical Data Mining	3
D601	Data Storytelling for Diverse Audiences	3
D602	Deployment	3
D603	Machine Learning	3
D604	Advanced Analytics	3
D605	Optimization	3
D606	Data Science Capstone	3

Total credits: 32

JOB MARKET OPPORTUNITY

By April 30, 2027, the job market for data analytics professionals is projected to remain strong, driven by continued growth in AI adoption, digital transformation, and data-centric decision-making across industries.

Employers in Maryland—especially healthcare, government, logistics, and tech—are expected to maintain high demand for analysts who can translate business problems into data-driven insights. With a completed master's degree and a solid portfolio, you can competitively target roles in the **\$90,000–\$120,000** range, with pathways into analytics engineering and senior analyst positions reaching **\$130,000–\$150,000** as you gain experience.

TIMELINE & CALENDAR

PLANNED START DATE: **NOVEMBER 1, 2026**

EXPECTED COMPLETION DATE: **APRIL 30, 2027**



This calendar provides a clear, structured overview of key milestones, deadlines, and planned activities. It translates the broader timeline into an organized visual, allowing stakeholders to track progress, anticipate upcoming priorities, and remain aligned with the overall roadmap. By pairing dates with defined actions, the calendar supports consistent pacing and informed decision-making throughout the project.

NOVEMBER

	COURSE
11/01	The Data Analytics Journey
11/08	The Data Analytics Journey
11/15	Data Storytelling
11/22	Data Storytelling
11/29	Data Management

DECEMBER

	COURSE
12/06	Data Management
12/13	Data Management
12/20	Analytics Programming
12/27	Analytics Programming

JANUARY

	COURSE
01/03	Analytics Programming
01/10	Data Prep & Exploration
01/17	Data Prep & Exploration
01/24	Data Prep & Exploration
01/31	Statistical Data Mining

FEBRUARY

	COURSE
02/07	Statistical Data Mining
02/14	Machine Learning
02/21	Machine Learning
02/28	Machine Learning

MARCH

	COURSE
03/07	Deployment
03/14	Deployment
03/21	Optimization
03/28	Optimization

APRIL

	COURSE
04/04	Advanced Analytics
04/11	Capstone
04/18	Capstone
04/25	Capstone

FINANCIAL PLAN

TUITION

Tuition	\$4,425.00
E-Books & Resource Fees	\$200.00
Total	\$4,625.00

FINANCIAL AID

Employer Assistance (2026)	\$1,250.00
Employer Assistance (2027)	\$2,500.00
Scholarships	\$0
Discounts	\$0
Other	\$0
Total	\$3,750.00

OUT-OF-POCKET COST

Tuition	\$4,625.00
Financial Aid	-\$3,750.00
Total	\$ 875.00

TUITION:

\$4,625

AID:

\$3,750

GAP:

- \$875

SCHOLARSHIP TRACKER

The scholarship tracking table provides a clear, organized view of all funding opportunities under consideration. It allows readers to monitor application statuses, deadlines, award amounts, and key requirements in one structured place. By centralizing this information, the table supports informed planning and ensures that each opportunity is evaluated and pursued with intention.

Scholarship	Award Amount	Awarding Body	Status
Rise Above Grant	Up to \$4,000	WGU	Applied
WGU AI Edge Scholarship	Up to \$5,000	WGU	Applied
Maryland Rising Together Scholarship	Up to \$4,000	WGU	Applied
Back to School Scholarship	Up to \$3,000	WGU	Applied
WGU Tech is Everywhere Scholarship	Up to \$3,000	WGU	Applied
Master Your Future Scholarship	Up to \$3,000	WGU	Applied
The University of You Scholarship	Up to \$10,000	WGU	Applied
WGU Leadership Scholarship	Up to \$5,000	WGU	Applied
Alumni Master's Scholarship	Up to \$3,000	WGU	Applied

COURSE DESCRIPTIONS

D596 THE DATA ANALYTICS JOURNEY (2 CREDITS)

Analytics is the creative use of data and statistical modeling to tell a compelling story that not only drives strategic action but also results in business value. The Data Analytics Journey uses the analytics life cycle to conceptualize the processes, tools, and techniques for implementing data analysis, data engineering, and analytics product management. Learners gain fluency in gathering requirements, asking business questions, establishing evaluation metrics, identifying communication models, and aligning the analytics project outcomes to business goals. It presents an overview of the various tracks offered in the program and the career options in these specializations.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner applies the data analytics life cycle to real world business scenarios.
- The learner proposes a data analytics plan to solve a business problem.
- The learner differentiates the variety of careers and roles related to the broad field of data analytics.
- The learner identifies potential career goals based on foundational knowledge of self and career pathways.

D597 DATA MANAGEMENT (3 CREDITS)

Data Management builds proficiency in using both relational and non-relational databases. Topics include selection of a data storage architecture, data types, data structures, normalization and denormalization, and querying databases. Structured Query Language (SQL) topics including Data Definition Language (DDL) and Data Manipulation Language (DML) are covered, including joins, aggregations, and transactions. Non-relational approaches to organizing and querying data are contrasted with relational approaches to build competency in adapting data storage architectures to business needs.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner recommends an appropriate data architecture.
- The learner examines the data available for analysis to determine their dimension, quality, relations, and limitations.
- The learner constructs a logical data model.
- The learner implements physical data models by incorporating required data.
- The learner performs database queries to answer a business question.
- The learner uses appropriate data manipulation language.

D598 ANALYTICS PROGRAMMING (3 CREDITS)

Analytics Programming builds algorithmic thinking using both the Python and R programming languages. This course builds from the foundations of programming. Learners use libraries and packages to perform common analytics tasks, including acquiring, organizing, and manipulating datasets. The course also presents methods for applying statistical functions and graphical user interfaces to perform basic analysis and to present findings.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner documents the required component parts of a complex programming task.
- The learner integrates programming language syntax, control elements, and modular components from common data analytics programming languages to create simple programs.
- The learner performs data acquisition and organization tasks using a data analytics programming language.

D599 DATA PREPARATION AND EXPLORATION (3 CREDITS)

Data Preparation and Exploration applies analytical programming skills to the early steps of the data analytics life cycle. This course covers cleaning data to ensure the structure, accuracy, and quality of the data; interpretation of descriptive and inferential statistics as well as visualizations of data; and wrangling data to prepare it for further analysis. The course introduces hypothesis testing, focusing on application for parametric tests, and addresses communication skills and tools to explain an analyst's findings to others within an organization.

The following courses are prerequisites: The Data Analytics Journey, Data Management, and Analytics Programming.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner interprets a data dictionary to understand the data set.
- The learner interprets probability, descriptive and inferential statistics, and visualization.
- The learner wrangles data to ensure accuracy, format, and integrity relevant to the task being performed.

D600 STATISTICAL DATA MINING (3 CREDITS)

Statistical Data Mining focuses on concepts in data preparation and supervised and unsupervised machine learning techniques. The course helps students gain basic knowledge in statistics, data preparation, regression, and dimensional reduction. Learners implement supervised models—specifically classification and prediction data mining models—to unearth relationships among variables that are not apparent with more surface-level techniques. The course also explains when, how, and why to use unsupervised models to best meet organizational needs.

The following course is prerequisite: Data Preparation and Exploration.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner performs linear and logistic regressions to make recommendations based on the results.
- The learner performs principal component analysis to make recommendations based on the results.

D601 DATA STORYTELLING FOR DIVERSE AUDIENCES (3 CREDITS)

Data Storytelling for Diverse Audiences focuses on communicating observations and patterns to diverse stakeholders, a key aspect of the data analytics life cycle. This course helps learners gain communication and storytelling skills in order to motivate change and answer business problems. It also covers data visualizations, audio representations, interactive dashboards, interpersonal communication, and presentation skills.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner demonstrates how interpersonal skills are applied to effectively communicate.
- The learner conducts an audience analysis.
- The learner communicates data insights to technical and nontechnical audiences.
- The learner creates data representations to offer insight into an organizational problem.
- The learner designs interactive dashboards to support executive decision-making.

D602 DEPLOYMENT (3 CREDITS)

Deployment is the practice of operationalizing data analysis within a business environment. Given an analysis, learners determine the business functional and non-functional requirements for wider use and implement pipelines and functions to deploy analyses at scale. Topics including security, scalability, usability, and availability are discussed.

Prerequisites for this course are Analytical Programming, Data Management, Data Preparation, and Statistical Data Mining.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner analyzes a business case to determine the requirements necessary for deployment.
- The learner implements a data product pipeline to address organizational needs.
- The learner implements a function to call and receive information between multiple systems for deployment.
- The learner deploys a data product based on project requirements.

D603 MACHINE LEARNING (3 CREDITS)

Machine Learning is the broad discipline of developing algorithms and statistical models to predict, classify, or cluster data and that iteratively improve over time. Machine Learning focuses on building, training, running, and testing supervised and unsupervised models and quantifying the accuracy and precision of those models to determine which may best be used in a particular business situation. Supervised methods covered include k-nearest neighbors, logistic regression, decision trees, and support vector machines. Unsupervised models covered include k-means clustering, hierarchical clustering, and t-distributed stochastic neighbor embedding (t-SNE). Ensemble methods are also presented.

Prerequisites are Analytics Programming and Statistical Data Mining.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner recommends a supervised machine learning model based on a comparison of model performance given a business problem.
- The learner recommends an unsupervised machine learning model based on a comparison of model performance given a business problem.
- The learner applies time series models in generating forecasts.

D604 ADVANCED ANALYTICS (3 CREDITS)

Advanced Analytics extends analytics techniques from machine learning to artificial intelligence more broadly, including topics in neural networks, deep learning, and natural language processing. The course covers approaches to developing these models, including PyTorch and TensorFlow. Students learn to apply a combination of techniques to solve complex business challenges, including computer vision and sentiment analysis.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner applies neural networks to solve a business problem.
- The learner applies natural language processing to solve a business problem.

D605 OPTIMIZATION (3 CREDITS)

Optimization is a large class of business problems requiring the iterative algorithmic maximization or minimization of one or more variables. Students in this course will select and use a variety of optimization approaches to address various business needs. The course covers classes of optimization problems at a foundational level (continuous/discrete, linear/nonlinear, and bounded/unbounded) and the solving of linear optimization problems in both Python and R through the use of gradient and non-gradient-based algorithms.

Analytics Programming is a prerequisite.

This course covers the following competencies:

- Begin your course by discussing your course planning tool report with your instructor and creating your personalized course plan together.
- The learner analyzes a business case to recommend a particular optimization approach.
- The learner identifies the objective function and constraints for an optimization problem.
- The learner solves an optimization problem programmatically.

D606 DATA SCIENCE CAPSTONE (3 CREDITS)

The Data Science Capstone integrates the learning in the MSDA core and the three courses within the specialization. The student evaluates various needs and opportunities in an organization or marketplace; identifies the business requirements; translates the business requirements into technical requirements; and creates a comprehensive project plan to solve the problem in a way that satisfies the customer or business needs. Projects within this specialization include the design and construction of machine learning approaches, optimization, and/or advanced analytics techniques as the project requires.

This course covers the following competencies:

- The learner proposes a data analytics approach to solve a business problem.
- The learner implements a data analytics approach to address a business problem.
- The learner communicates information and data trends using data storytelling and visualization.



IMPORTANT CONTACTS

- **Enrollment Advisor:** Anthony Scott,
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anthony.scott@wgu.edu
- **Scholarship Department:**
scholarships@wgu.edu,
wgu.scholarshipuniverse.com
- **Program Mentor:** [TBD]
- **Career Services:**
careers@wgu.edu, careers.wgu.edu
- **Professional Mentor:** [Open]
- **Academic Coaching:**
tinyurl.com/wgucoach
- **Student Services:** (877) 435-7948

RISK ASSESSMENT

Potential challenges and mitigation strategies:

- Spikes in workload around Mid-year review time (April)
- Need for more personal time during holidays (November to December)
- Unpredictable changes in work schedule

EXIT STRATEGY

My exit strategy focuses on converting the technical, analytical, and business competencies gained throughout the MSDA program into immediate, market-ready value. By the time I complete the degree on **April 30, 2027**, I will have a full portfolio of applied analytics projects, hands-on experience with SQL, Python, machine learning, data storytelling, and deployment pipelines, and a capstone demonstrating end-to-end problem solving. My goal is to transition directly into a **data analyst, business analyst, or analytics engineer** role within 30–90 days of graduation.

Throughout the program, I will build a structured portfolio that includes:

- A business problem analysis and analytics plan (D596)
- SQL-based data modeling and querying projects (D597)
- Python/R data acquisition and manipulation scripts (D598)
- A full data cleaning, wrangling, and statistical exploration project (D599)
- Regression, PCA, and supervised/unsupervised ML models (D600–D603)
- A deployed data pipeline or analytics function (D602)
- A deep learning or NLP project (D604)
- An optimization solution demonstrating algorithmic decision-making (D605)
- A comprehensive capstone integrating business requirements, modeling, and storytelling (D606)

By graduation, I will have a complete set of artifacts that demonstrate both technical depth and business impact. My exit strategy includes updating my résumé, LinkedIn profile, and professional website with these projects, targeting roles that align with my background in operations, IT management, and analytics. I will begin applying for roles during the final month of the program, leveraging my portfolio, capstone, and new credentials to secure interviews quickly. My objective is to secure a role in the **\$90,000–\$120,000** range, with a pathway toward analytics engineering or senior analyst positions as I gain experience.

MILESTONES & ROADMAP

This milestone visual outlines the critical checkpoints that will guide my progress from the start of the MSDA program through graduation and into my transition into a data-focused career. Each milestone represents a concrete achievement—academic, financial, professional, or portfolio-based—that ensures I remain on track, maintain momentum, and convert my learning into measurable outcomes. Together, these milestones form a clear roadmap that keeps my goals visible, my timeline structured, and my post-graduation success intentional.

12/31/2026:

Secure first tuition reimbursement
from employer

01/15/2027:

Complete First Portfolio Project and
add it to GitHub

02/15/2027:

Build Programming & Data Wrangling
Portfolio Project

03/10/2027:

Publish First Machine Learning Model
on GitHub

03/31/2027:

Deploy a Working Data Pipeline or
Analytics Function

04/10/2027:

Complete Advanced Analytics Project
(NLP or Neural Network)

04/30/2027:

Deliver and Publish Capstone Project

07/31/2027:

Secure a Data Analytics Role

POST-GRADUATION 90-DAY PLAN

FINALIZE PROFESSIONAL BRANDING

MAY 1, 2027 TO MAY 15, 2027

Update all professional materials to reflect your new degree and portfolio:

- ❑ Refresh your résumé with MSDA completion, capstone, and technical skills
- ❑ Update LinkedIn with your new credential, projects, and keywords
- ❑ Publish your portfolio projects on GitHub and link them to your profile
- ❑ Add your capstone as a featured project with a clear business problem, methodology, and results This ensures recruiters immediately see your readiness for analytics roles.

BEGIN TARGETED JOB SEARCH

MAY 15, 2027 TO JUNE 30, 2027

Start applying to roles that align with your background and new skills:

- **Data Analyst**
- **Business Analyst**
- **Operations Analyst**
- **Analytics Engineer**

Focus on Maryland/DC employers in healthcare, government contracting, logistics, and tech — all strong analytics markets.

Focus on these job boards:

- [LinkedIn](#)
- [Indeed](#)
- [Analytics Engineering Jobs](#)
- [Wellfound](#)
- [Built In](#)
- [USAJOBS](#)
- [GovernmentJobs](#)
- [HigherEdJobs](#)

Aim for **10–15 targeted applications per week**.

STRENGTHEN TECHNICAL INTERVIEW READINESS

MAY 2027 TO JUNE 2027

Prepare for analytics interviews by practicing:

- ☐ SQL challenges (joins, aggregations, window functions)
- ☐ Python data manipulation (Pandas, NumPy)
- ☐ Exploratory data analysis walkthroughs
- ☐ Machine learning model explanations (regression, classification, clustering)
- ☐ Dashboard storytelling using Power BI or Tableau

This ensures you can confidently demonstrate both technical and business thinking.

EXPAND PROFESSIONAL NETWORK

MAY 2027 TO JULY 2027

Engage with analytics communities to increase visibility:

- ☐ Attend local meetups in Baltimore/DC
- ☐ Join data analytics Slack groups and LinkedIn communities
- ☐ Connect with hiring managers and analysts in target companies
- ☐ Share projects publicly once per week Networking accelerates recruiter outreach and interview opportunities.

BEGIN INTERVIEW CYCLE

JUNE 2027 TO JULY 2027

Expect:

- ☐ 2–4 recruiter screens
- ☐ 2–3 technical interviews
- ☐ 1–2 case studies or take-home assignments Your portfolio will significantly increase your pass-through rate.

SECURE A DATA ANALYTICS ROLE

BY JULY 2027

BEGIN ONBOARDING & SKILL DEEPENING

AUGUST 2027

Once hired, continue strengthening analytics skill set:

- ☐ Learn your employer's data stack (Snowflake, Databricks, AWS, etc.)
- ☐ Build internal dashboards and pipelines
- ☐ Identify opportunities for early wins This sets you up for strong performance reviews and rapid advancement.

CLOSING STATEMENT

This learning plan represents my commitment to completing the MS in Data Analytics program with clarity, discipline, and purpose. By following the structured timeline, financial strategy, milestone roadmap, and risk mitigation steps outlined above, I will graduate on April 30, 2027, fully prepared to transition into a data-focused career. Each course in this program builds toward a portfolio of applied analytics work that demonstrates both technical capability and business impact.

As I progress through this journey, I will remain focused on producing high-quality projects, strengthening my professional brand, and aligning my skills with the needs of the current job market. Upon graduation, I will leverage my completed degree, published portfolio, and capstone project to secure a data analytics role by July 31, 2027. This plan reflects not only the steps required to complete the program, but also the strategic actions necessary to convert my education into meaningful career advancement.

With this roadmap in place, I am positioned to complete the program successfully, grow as a data professional, and enter the next phase of my career with confidence and momentum.



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Turning Data into Decisions
